TASK SPECIFICATION: P3-03 PHASE P3 — FRONTEND ARCHITECTURE (REACT 19 + TYPESCRIPT)

Typed API Client, Auth Context & Route Guards

Git Branch: `feature/P3-03-api-client-auth-context` · Target Pull Request: feature/P3-03-api-client-auth-context

Core Objective & Technical Scope:
Typed HTTP client with Axios, TanStack Query integration, AuthContext provider, automated JWT token refresh interceptors, and protected route guards.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P3-03-api-client-auth-context`
Upstream Dependencies	P3-01, P2-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Create configured Axios client with automatic Authorization header injection.
2. Implement token refresh interceptor on 401 responses with queued request replay.
3. Build AuthProvider managing user session, roles, and login/logout state.
4. Implement ProtectedRoute and RoleGuard components for declarative authorization.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/shared/http/*`
- `frontend/src/features/auth/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- 401 response triggers token refresh once then logs out on failure.
- Unauthorized access attempts redirect cleanly to login with return path preserved.

VERIFICATION & TEST COVERAGE STRATEGY

- HTTP interceptor tests
- MSW mock integration tests
- Route guard tests